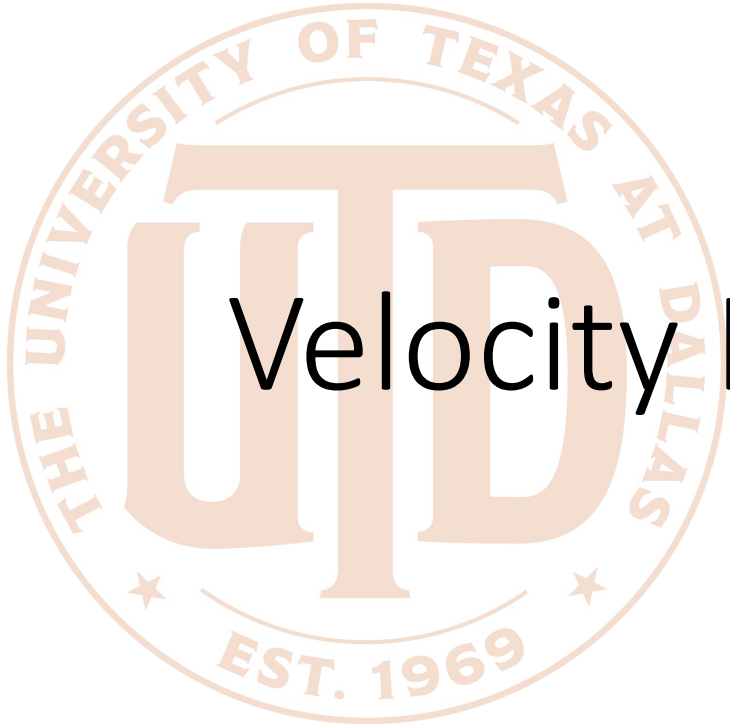




# Velocity Kinematics: Angular Velocity and Linear Velocity

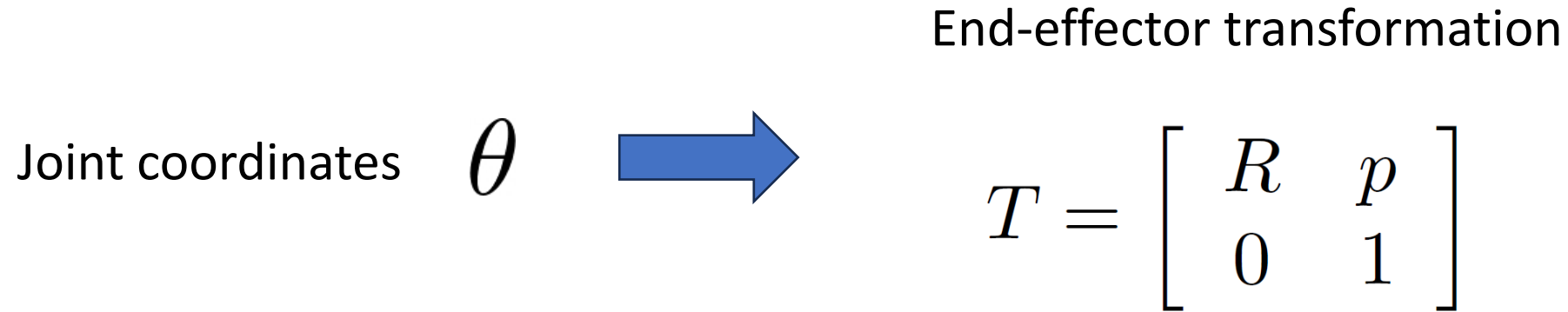
CS 6341 Robotics

Instructor Yu Xiang

The University of Texas at Dallas

# Forward Kinematics

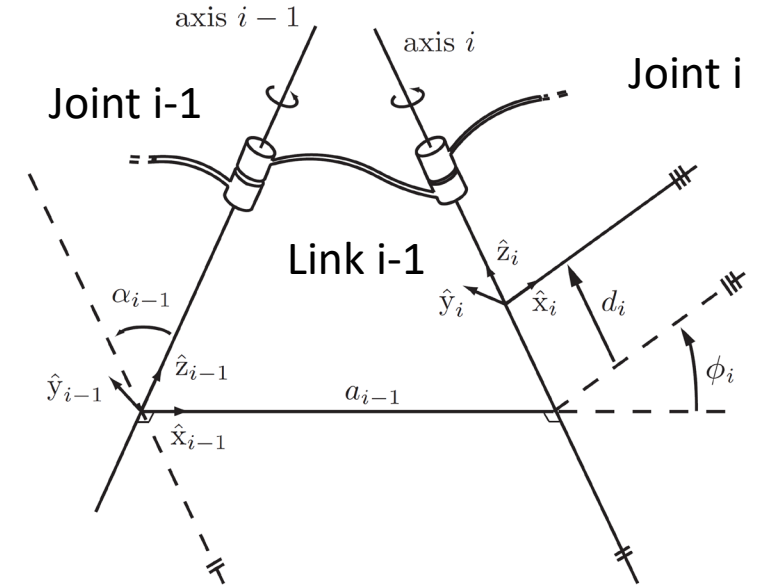
- Forward kinematics of a robot: calculation of the position and orientation of its end-effector from its joint coordinates



# Forward Kinematics with D-H Parameters

- Link frame transformation

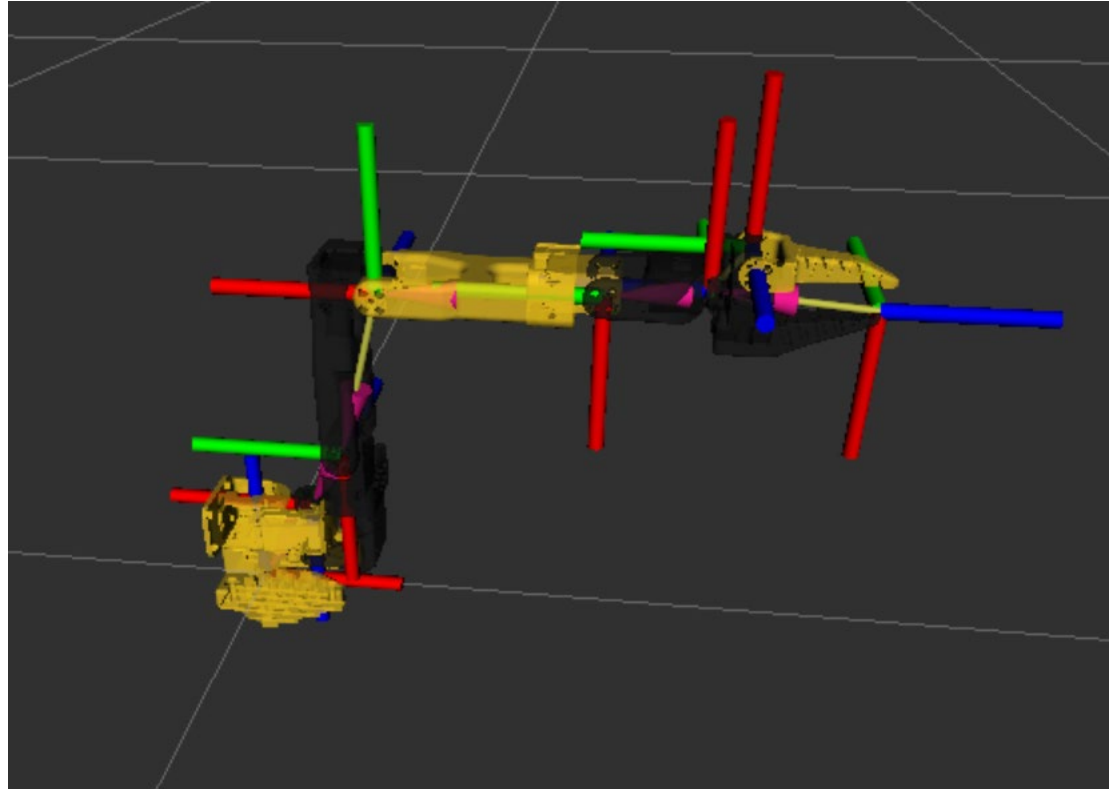
$$T_{i-1,i} = \text{Rot}(\hat{x}, \alpha_{i-1}) \text{Trans}(\hat{x}, a_{i-1}) \text{Trans}(\hat{z}, d_i) \text{Rot}(\hat{z}, \phi_i)$$



$$T_{0n}(\theta_1, \dots, \theta_n) = T_{01}(\theta_1) T_{12}(\theta_2) \cdots T_{n-1,n}(\theta_n)$$

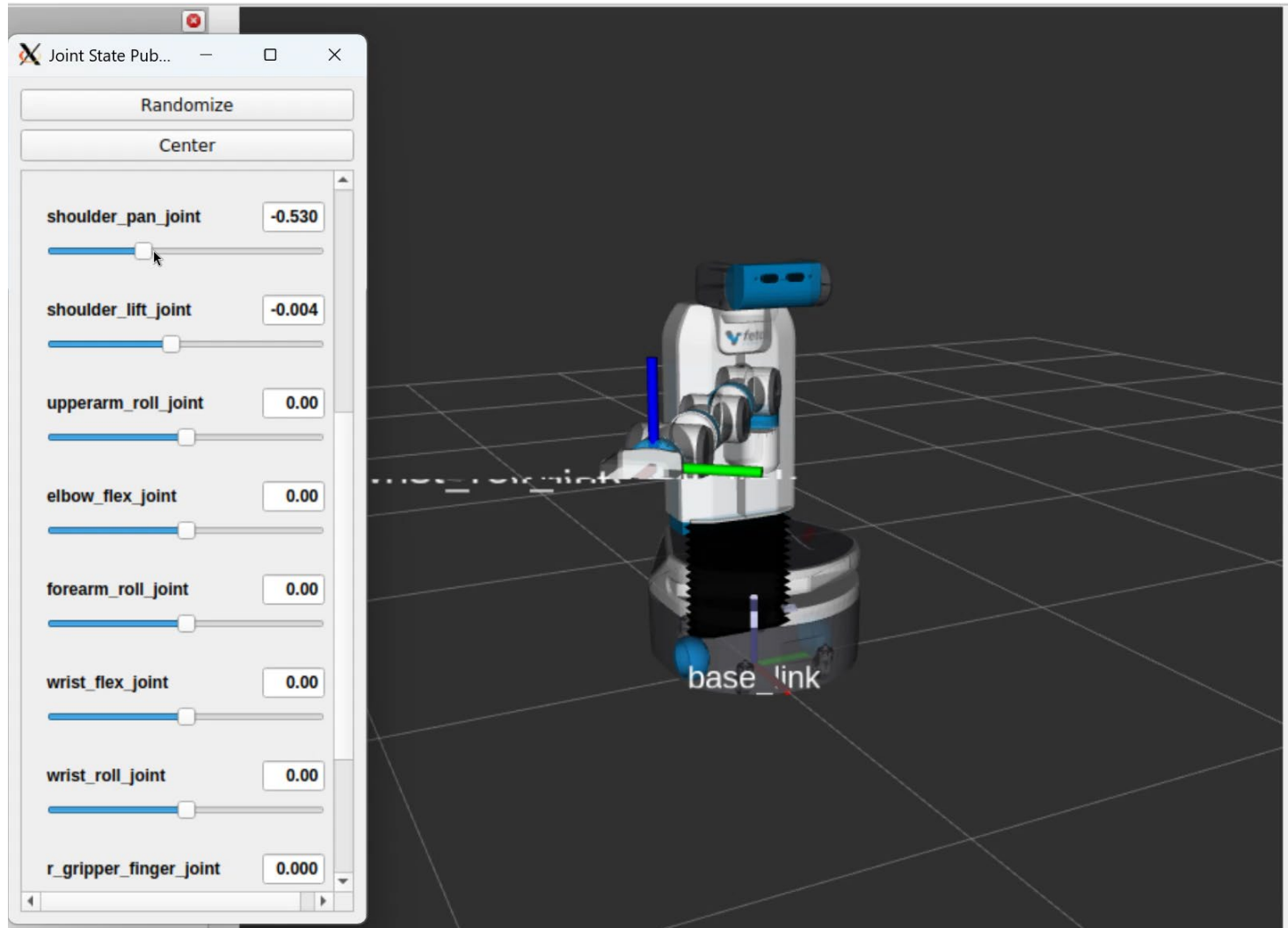
# Forward Kinematics

SO101 Arm



$$T_{0n}(\theta_1, \dots, \theta_n) = T_{01}(\theta_1)T_{12}(\theta_2) \cdots T_{n-1,n}(\theta_n)$$

# What is the Velocity of the End-effector?



# Why we need to care about end-effector velocity?



<https://www.youtube.com/watch?v=wXxrmussq4E>

# Velocity Kinematics

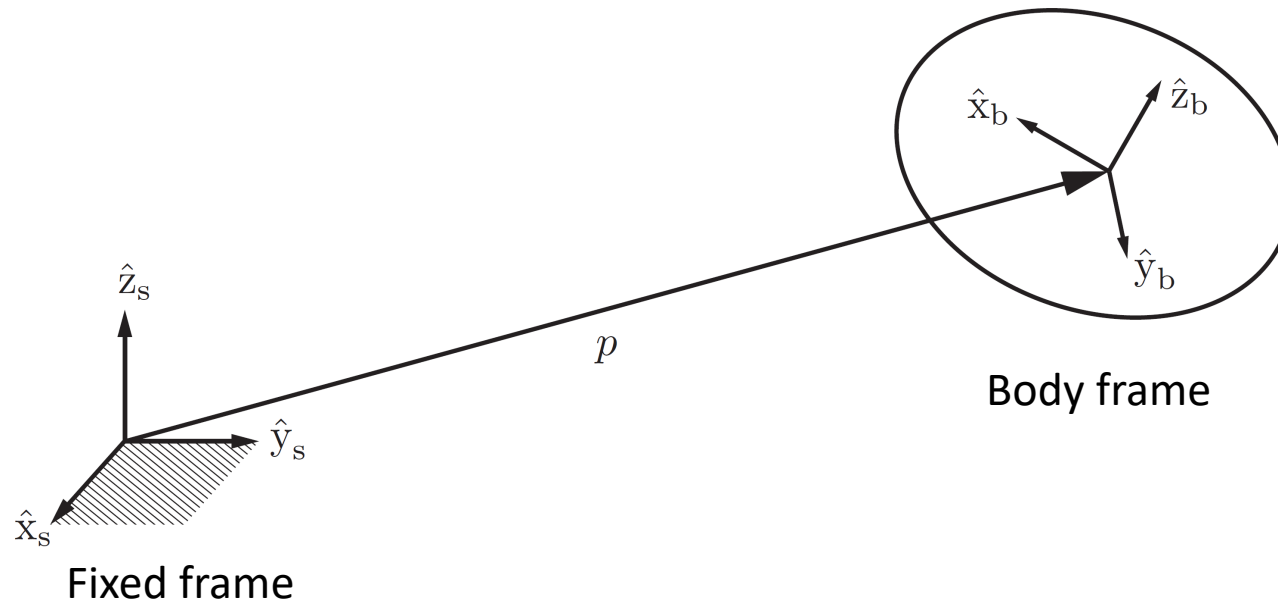
- Given joint positions and velocities  $\theta \in \mathbb{R}^n$   $\dot{\theta}$
- Compute the velocity of the end-effector

End-effector configuration

$$T = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix} \xrightarrow[\dot{p}(t) = \frac{d}{dt}p(t)]{\dot{R}(t) = \frac{d}{dt}R(t)} \dot{T} = \begin{bmatrix} \dot{R} & \dot{p} \\ 0 & 0 \end{bmatrix}$$

What is this?

# Recall Rigid-Body in 3D



- Origin of the body frame

$$p = p_1 \hat{x}_s + p_2 \hat{y}_s + p_3 \hat{z}_s$$

- Axes of the body frame

$$\hat{x}_b = r_{11} \hat{x}_s + r_{21} \hat{y}_s + r_{31} \hat{z}_s,$$

$$\hat{y}_b = r_{12} \hat{x}_s + r_{22} \hat{y}_s + r_{32} \hat{z}_s,$$

$$\hat{z}_b = r_{13} \hat{x}_s + r_{23} \hat{y}_s + r_{33} \hat{z}_s.$$

Translation

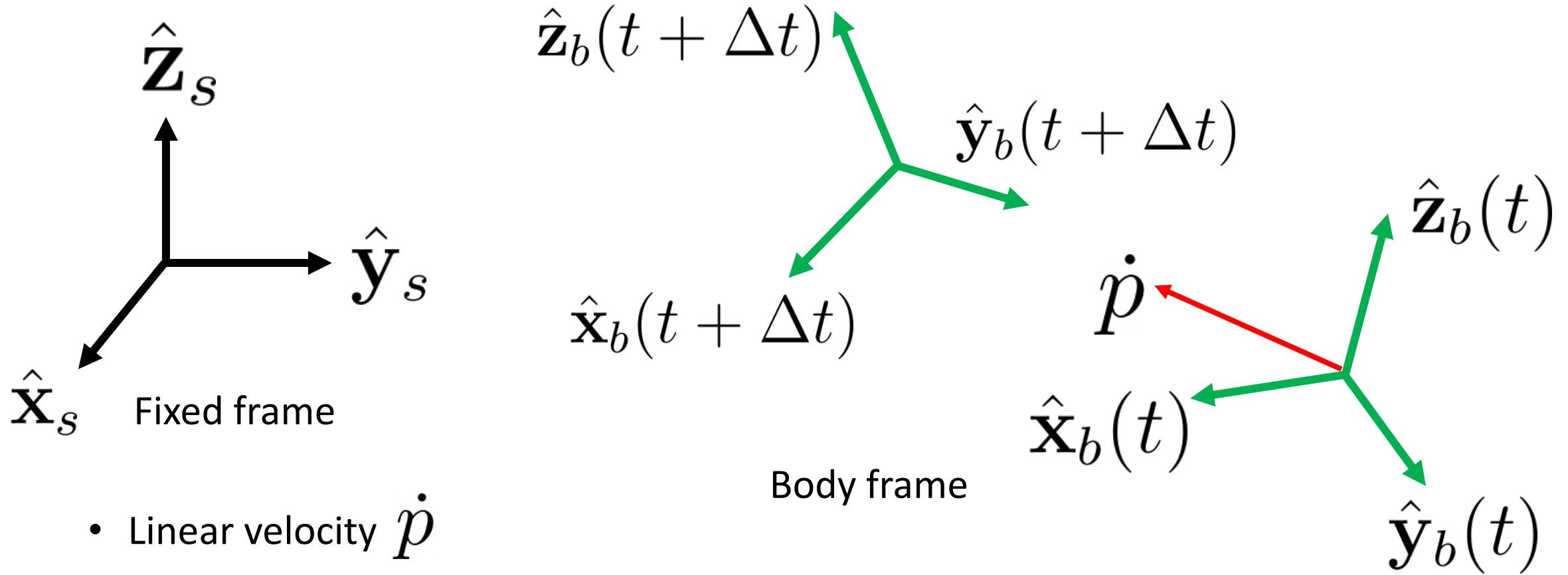
$$p = \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix}$$

Rotation matrix

$$R = [\hat{x}_b \ \hat{y}_b \ \hat{z}_b] = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

Meanings of the column vectors

# Angular Velocity and Linear Velocity



- Linear velocity  $\dot{\mathbf{p}}$

The linear velocity of the origin of {b} expressed in the fixed frame {s}

- How about  $\dot{\mathbf{R}}$  ?

# Recall Rotating a Vector or a Frame

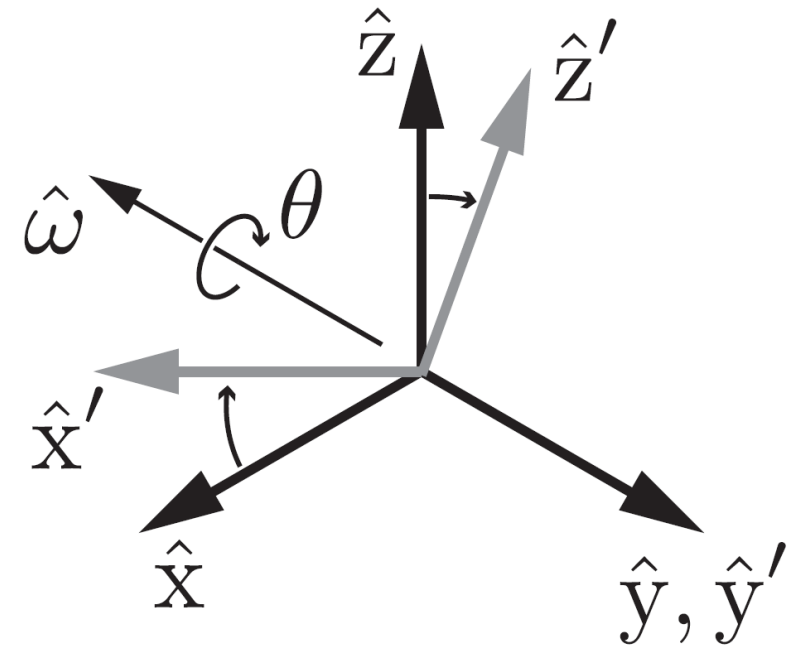
- Rotate frame  $\{c\}$  about a unit axis  $\hat{\omega}$  by  $\theta$  to get frame  $\{c'\}$ ,  $\{c\}$  is aligned with  $\{s\}$  in the beginning

$$R = R_{sc'}$$

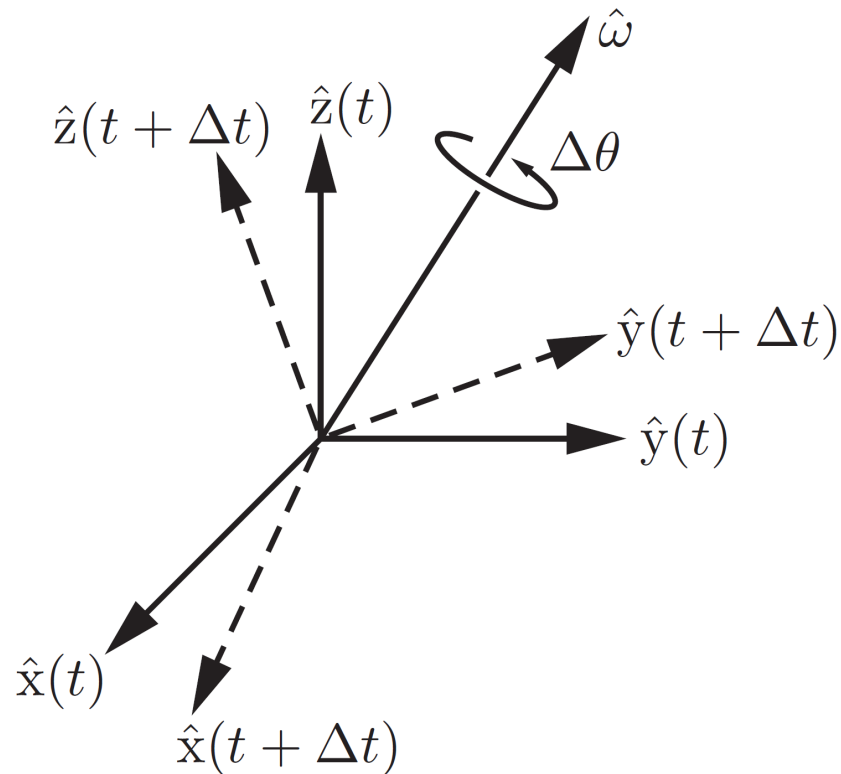
frame  $\{c'\}$  relative to frame  $\{s\}$

- Rotation operation

$$R = \text{Rot}(\hat{\omega}, \theta)$$



# Angular Velocities



- Axes  $\{\hat{x}, \hat{y}, \hat{z}\}$  Unit length

Rotating around  $\hat{\omega}$  by  $\Delta\theta$

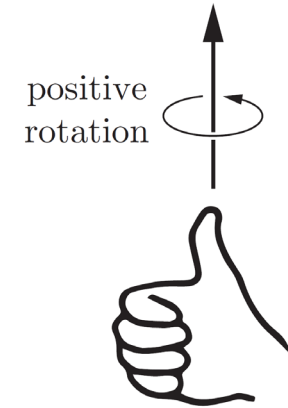
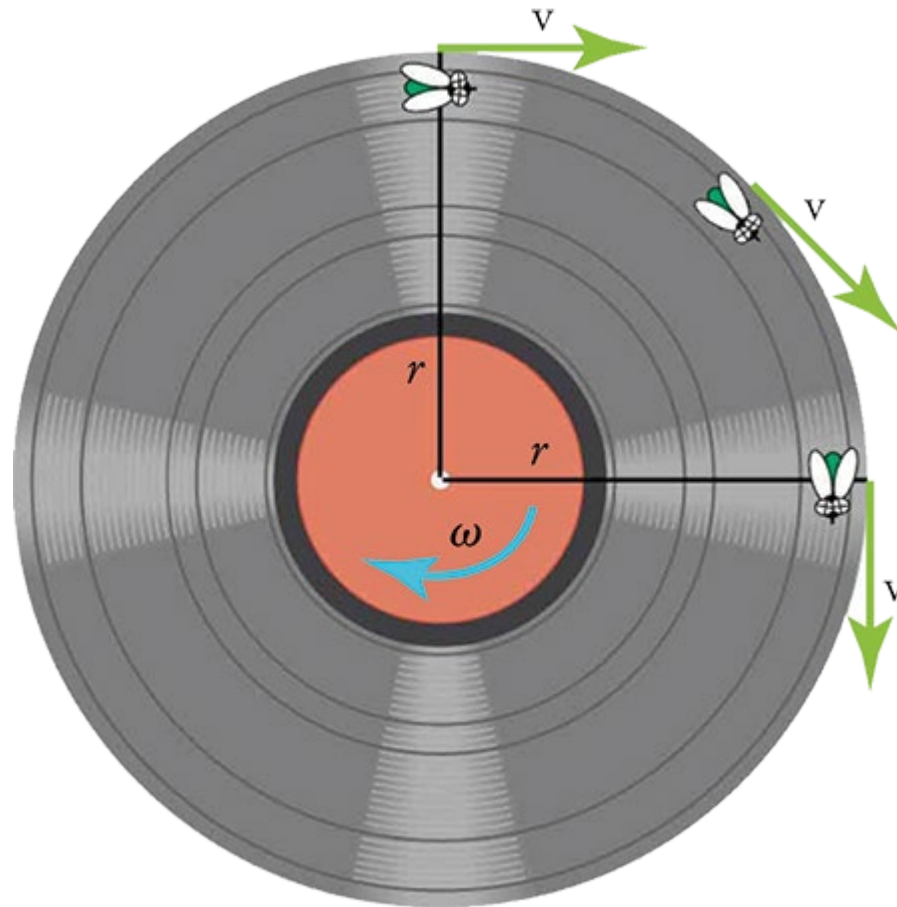
$\hat{\omega}$  is coordinate free for now

$\Delta t \rightarrow 0 \quad \Delta\theta / \Delta t \rightarrow \dot{\theta}$  **Instantaneous velocity**

$\hat{\omega}$  instantaneous axis of rotation

- **Definition** Angular velocity  $\omega = \hat{\omega} \dot{\theta}$

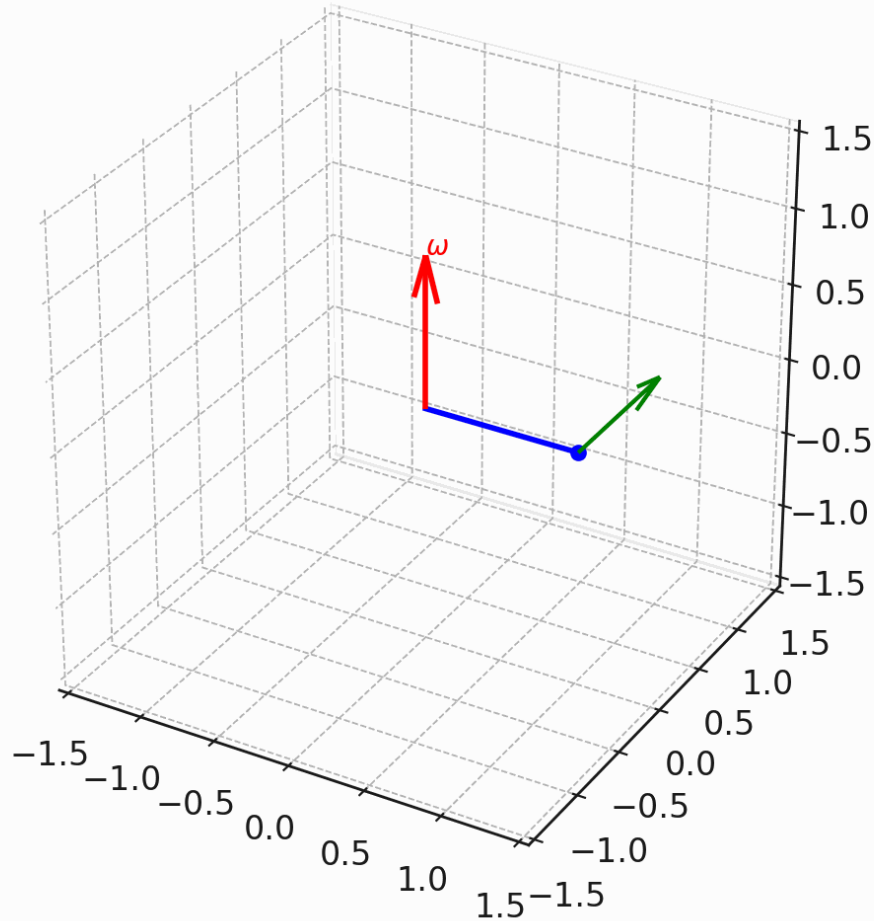
# Angular Velocity and Tangential Velocity



Speed  $v = r \omega$

<https://openstax.org/books/physics/pages/6-1-angle-of-rotation-and-angular-velocity>

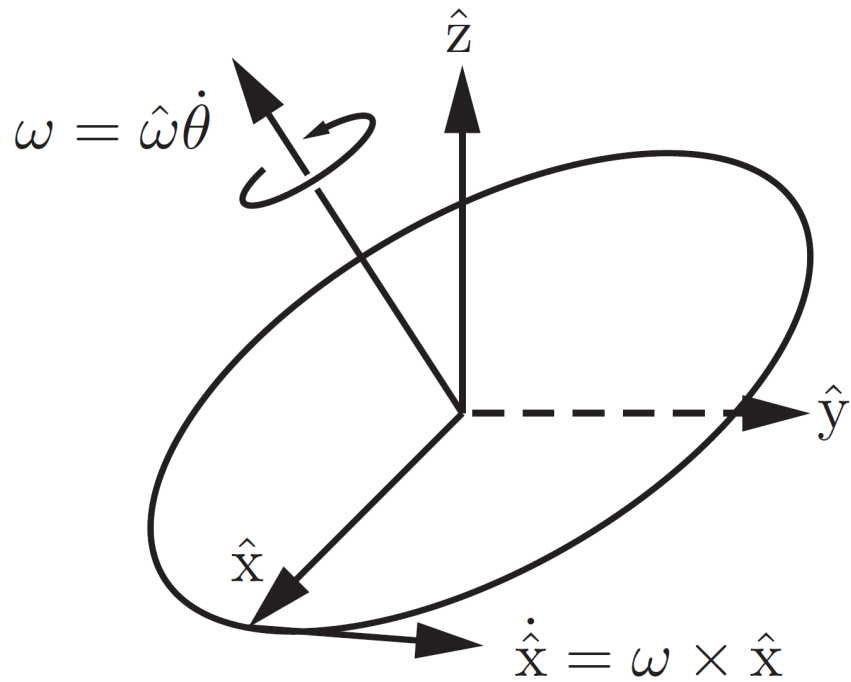
# Angular Velocities



Generated by ChatGPT

- **Red arrow**  $\rightarrow$  angular velocity vector  $\omega$
- **Blue line**  $\rightarrow$  rotating body-fixed axis.
- **Green arrow**  $\rightarrow$  instantaneous linear velocity of the blue endpoint ( $\mathbf{v} = \omega \times \mathbf{r}$ ).

# Angular Velocities

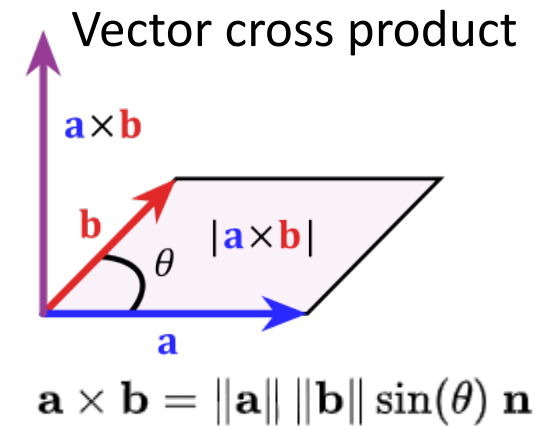


- Angular velocity  $\omega = \hat{\omega}\dot{\theta}$
- Compute time derivatives of these axes caused by rotation  $\dot{\hat{\mathbf{x}}}$  (tangential velocity)

$$\dot{\hat{\mathbf{x}}} = \omega \times \hat{\mathbf{x}}$$

$$\dot{\hat{\mathbf{y}}} = \omega \times \hat{\mathbf{y}}$$

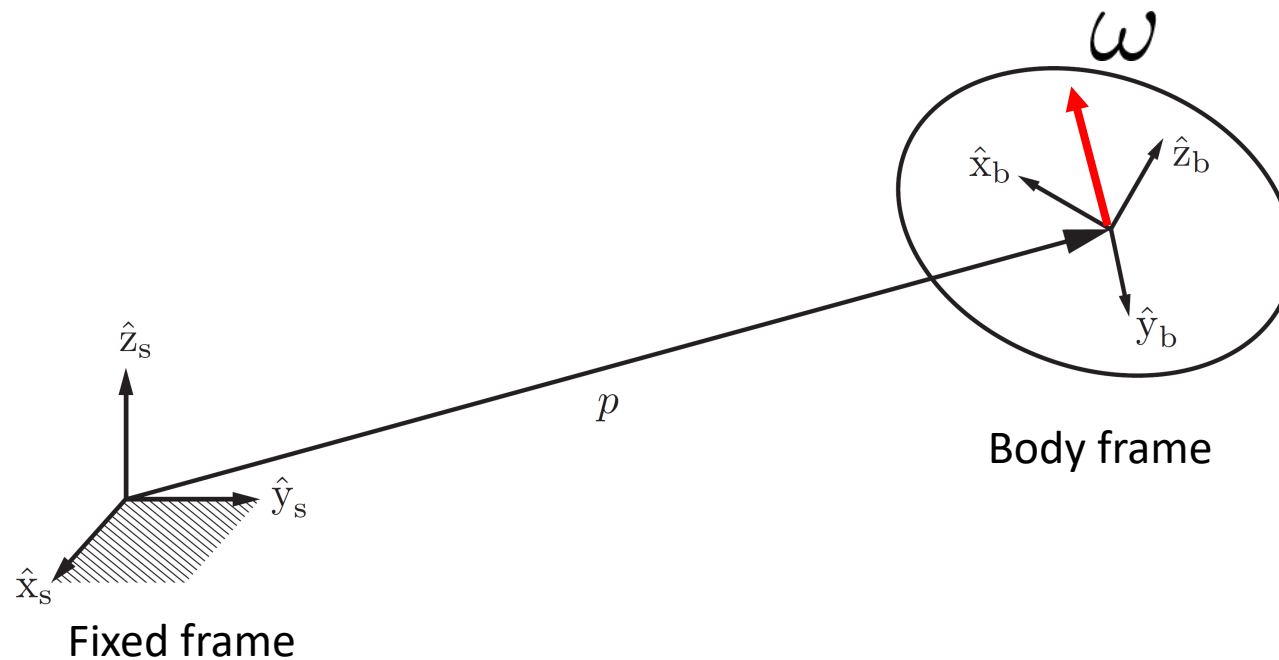
$$\dot{\hat{\mathbf{z}}} = \omega \times \hat{\mathbf{z}}$$



[https://en.wikipedia.org/wiki/Cross\\_product](https://en.wikipedia.org/wiki/Cross_product)

# Angular Velocities

- To express these equations in coordinates, we must choose a reference frame for  $\omega$ 
  - Two natural choices: fixed frame  $\{s\}$  or body frame  $\{b\}$



# Angular Velocities in Fixed Frame

- Consider fixed frame  $\{s\}$

- Orientation of the body frame at time  $t$   $R(t) = [\hat{x}_b \ \hat{y}_b \ \hat{z}_b]$   
 $= [r_1(t) \ r_2(t) \ r_3(t)]$
- Time rate of change  $\dot{R}(t)$

- Angular velocity  $\omega_s \in \mathbb{R}^3$

$$\dot{r}_i = \omega_s \times r_i, \quad i = 1, 2, 3.$$

Column

$$\begin{aligned} \dot{\hat{x}} &= \omega \times \hat{x}, \\ \dot{\hat{y}} &= \omega \times \hat{y}, \\ \dot{\hat{z}} &= \omega \times \hat{z}. \end{aligned}$$

$$\dot{R} = [\omega_s \times r_1 \ \omega_s \times r_2 \ \omega_s \times r_3] = \omega_s \times R.$$

# Skew-symmetric Matrix

[https://en.wikipedia.org/wiki/Skew-symmetric\\_matrix](https://en.wikipedia.org/wiki/Skew-symmetric_matrix)

$$x = [x_1 \ x_2 \ x_3]^T \in \mathbb{R}^3$$

$$[x] = \begin{bmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{bmatrix}$$

$$[x] = -[x]^T$$

$$\omega_s \times R = [\omega_s]R$$

$$\dot{R} = \omega_s \times R$$

$$[\omega_s]R = \dot{R}$$

$$[\omega_s] = \dot{R}R^{-1}$$

# Skew-symmetric Matrix

Proposition  $R[\omega]R^T = [R\omega] \quad \omega \in \mathbb{R}^3 \quad R \in SO(3)$

See Lynch & Park for proof

*Proof.* Letting  $r_i^T$  be the  $i$ th row of  $R$ , we have

$$\begin{aligned} R[\omega]R^T &= \begin{bmatrix} r_1^T(\omega \times r_1) & r_1^T(\omega \times r_2) & r_1^T(\omega \times r_3) \\ r_2^T(\omega \times r_1) & r_2^T(\omega \times r_2) & r_2^T(\omega \times r_3) \\ r_3^T(\omega \times r_1) & r_3^T(\omega \times r_2) & r_3^T(\omega \times r_3) \end{bmatrix} \\ &= \begin{bmatrix} 0 & -r_3^T\omega & r_2^T\omega \\ r_3^T\omega & 0 & -r_1^T\omega \\ -r_2^T\omega & r_1^T\omega & 0 \end{bmatrix} \\ &= [R\omega], \end{aligned}$$

# Angular Velocities in Body Frame

- Consider body frame  $\{b\}$   $\omega_b$

Change of reference frame  $\omega_s = R_{sb}\omega_b$

$$\omega_b = R_{sb}^{-1}\omega_s = R^{-1}\omega_s = R^T\omega_s$$

$$\begin{aligned} [\omega_b] &= [R^T\omega_s] \\ &= R^T[\omega_s]R \quad (\text{proposition}) & [\omega_s] &= \dot{R}R^{-1} \\ &= R^T(\dot{R}R^T)R \\ &= R^T\dot{R} = R^{-1}\dot{R} \end{aligned}$$

# Angular Velocities

- Orientation of the body frame at time  $t$  in the fixed frame  $R(t)$   
 $R_{sb}(t)$

- Angular velocity  $\omega$

$$\dot{R}R^{-1} = [\omega_s]$$

$$R^{-1}\dot{R} = [\omega_b]$$

- Change of reference frame of angular velocity

$$\omega_c = R_{cd}\omega_d$$

# Velocity Kinematics

- Compute the velocity of the end-effector

End-effector configuration

$$T = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix} \longrightarrow \dot{T} = \begin{bmatrix} \dot{R} & \dot{p} \\ 0 & 0 \end{bmatrix}$$

What is this?

- $\dot{p}$  linear velocity of the origin of {b} expressed in the fixed frame {s}

$$\dot{R} = [\omega_s]R \quad \dot{R} = R[\omega_b] \quad \text{Related to angular velocity}$$

- Velocity kinematics: how to compute linear velocity and angular velocity given joint positions and velocities? (future lectures)

# Summary

- Velocity Kinematics
- Linear Velocity
- Angular velocity

# Further Reading

- Chapter 3 in Kevin M. Lynch and Frank C. Park. Modern Robotics: Mechanics, Planning, and Control. 1st Edition, 2017